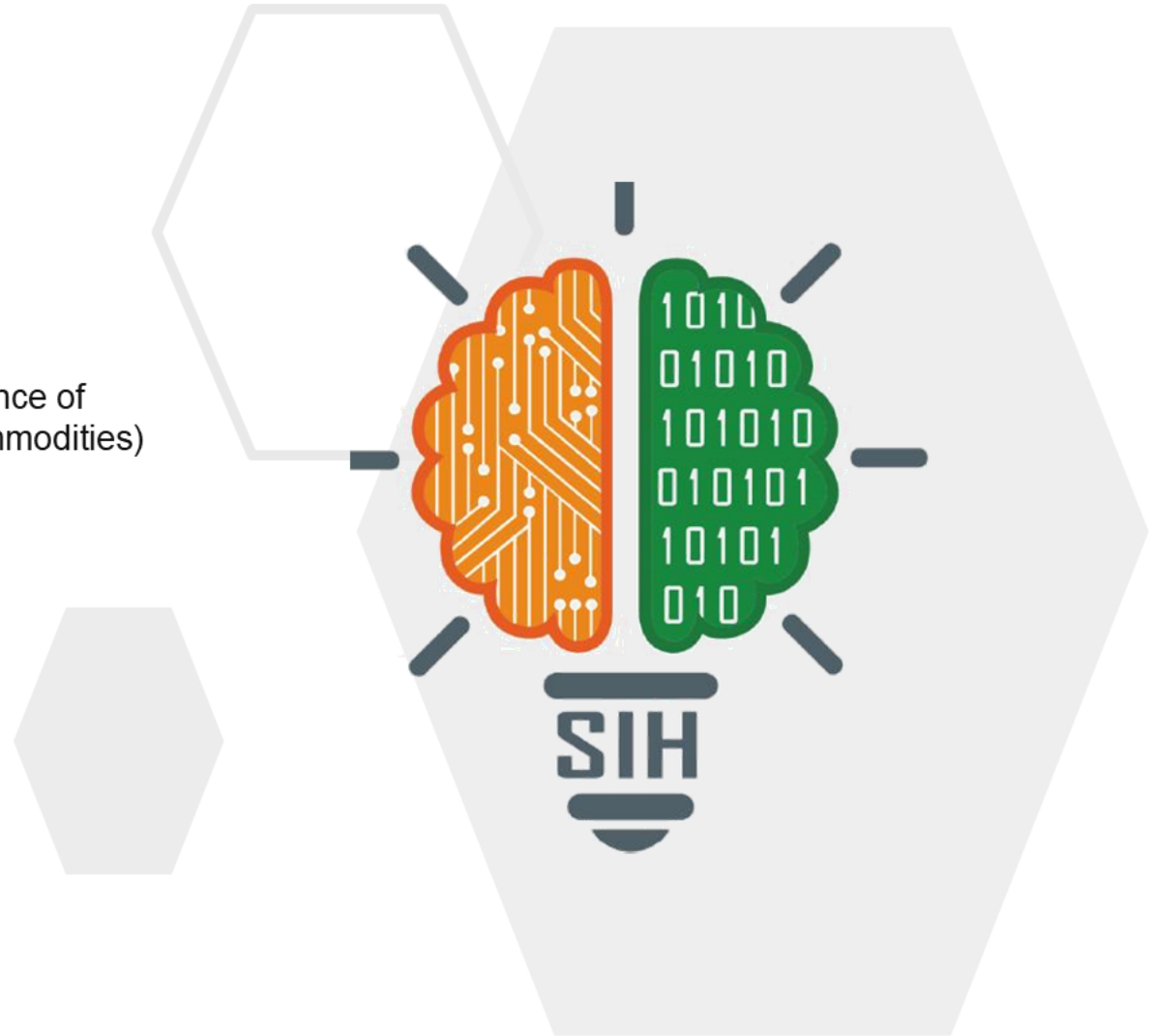


## NIYAMLENS

*Evidence-grade packaged commodity inspection*

- **Problem Statement ID** – SIH26034
- **Problem Statement Title** – Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels.
- **Theme** – Agriculture, FoodTech & Rural Development
- **PS Category** – Software
- **Team ID** – College allocation pending
- **Team Name** – NiyamLens

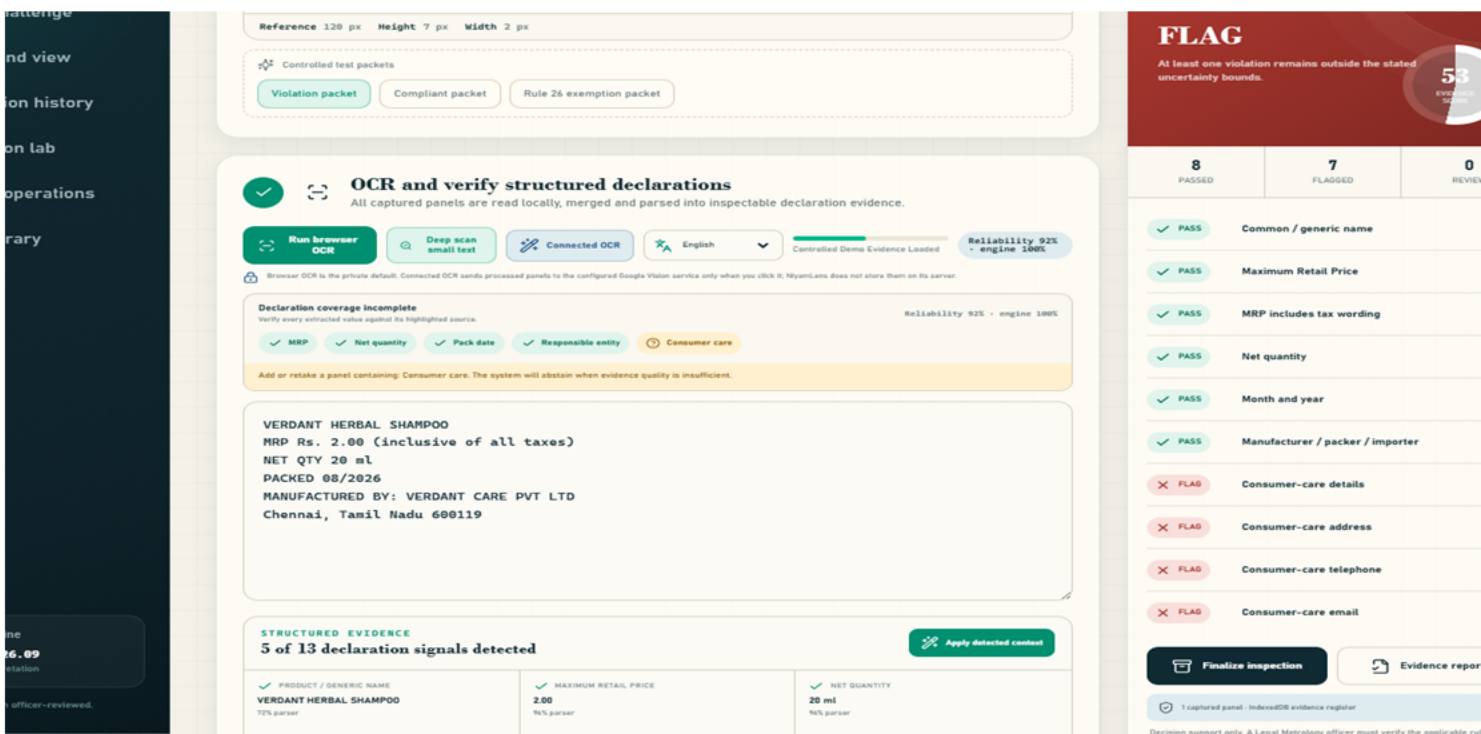


## ◆ Proposed Solution / Approach

### From image to defensible evidence

- **Guided capture:** Front, back, side and declaration panels; quality checks and original SHA-256.
- **Hybrid OCR:** Three-pass local scan, seven-pass deep scan and an explicit connected-OCR option.
- **Rules-as-code:** Rule 6 declarations, Rule 7 Table-I typography and Rule 26 exemptions.
- **Safe verdict:** PASS, FLAG, REVIEW or EXEMPT with source regions, uncertainty and officer audit.

OCR proposes evidence. Rules and officers decide.



The screenshot displays the NiyamLens interface. At the top, it shows dimensions: Reference 120 px, Height 7 px, Width 2 px. Below this are tabs for 'Controlled test packets', 'Violation packet', 'Compliant packet', and 'Rule 26 exemption packet'. The main section is titled 'OCR and verify structured declarations' with a sub-header 'All captured panels are read locally, merged and parsed into inspectable declaration evidence.' It includes buttons for 'Run browser OCR', 'Deep scan small test', 'Connected OCR', and a language dropdown set to 'English'. A progress bar shows 'Controlled Demo Evidence Loaded' and 'Reliability 92% - engine 100%'. A section titled 'Declaration coverage incomplete' lists 'MRP', 'Net quantity', 'Pack date', 'Responsible entity', and 'Consumer care'. Below this is a text block for 'VERDANT HERBAL SHAMPOO' with details like 'MRP Rs. 2.00 (inclusive of all taxes)', 'NET QTY 20 ml', 'PACKED 08/2026', and 'MANUFACTURED BY: VERDANT CARE PVT LTD Chennai, Tamil Nadu 600119'. A 'STRUCTURED EVIDENCE' section shows '5 of 13 declaration signals detected' with a table of evidence items. On the right, a 'FLAG' summary shows '8 PASSED', '7 FLAGGED', and '0 REVIEW'. Below this is a list of items with 'PASS' or 'FLAG' status, including 'Common / generic name', 'Maximum Retail Price', 'MRP includes tax wording', 'Net quantity', 'Month and year', 'Manufacturer / packer / importer', 'Consumer-care details', 'Consumer-care address', 'Consumer-care telephone', and 'Consumer-care email'. At the bottom right are buttons for 'Finalize inspection' and 'Evidence report'.

### WORKING PROTOTYPE: OCR + STRUCTURED EVIDENCE + RULE VERDICT



Decision support only; Legal Metrology officer verification remains explicit.

**ZONE 1 — BROWSER PWA**

- React 18 + Vite
- Canvas rectification
- Tesseract.js bundled OCR
- Installable offline service worker

**ZONE 2 — TRUST + RULE LAYER**

- SHA-256 evidence integrity
- IndexedDB local register
- Deterministic versioned matrix
- Rule outcome + officer disposition

**ZONE 3 — OPTIONAL SERVICE**

- Explicit connected-OCR action
- Vercel serverless function
- Google Vision key stays server-side
- Core flow remains complete offline

**COMPONENTS / TECHNOLOGY  
STACK**

React 18 • Vite • Canvas API • Tesseract.js • Web Crypto • IndexedDB • Playwright • Vercel

## Feasibility

- 31 implemented capabilities
- 54/54 automated tests pass
- Production build + browser QA pass

## Viability

- PWA + bundled OCR
- Local IndexedDB evidence
- No cloud required for core flow

## Practical Implementation

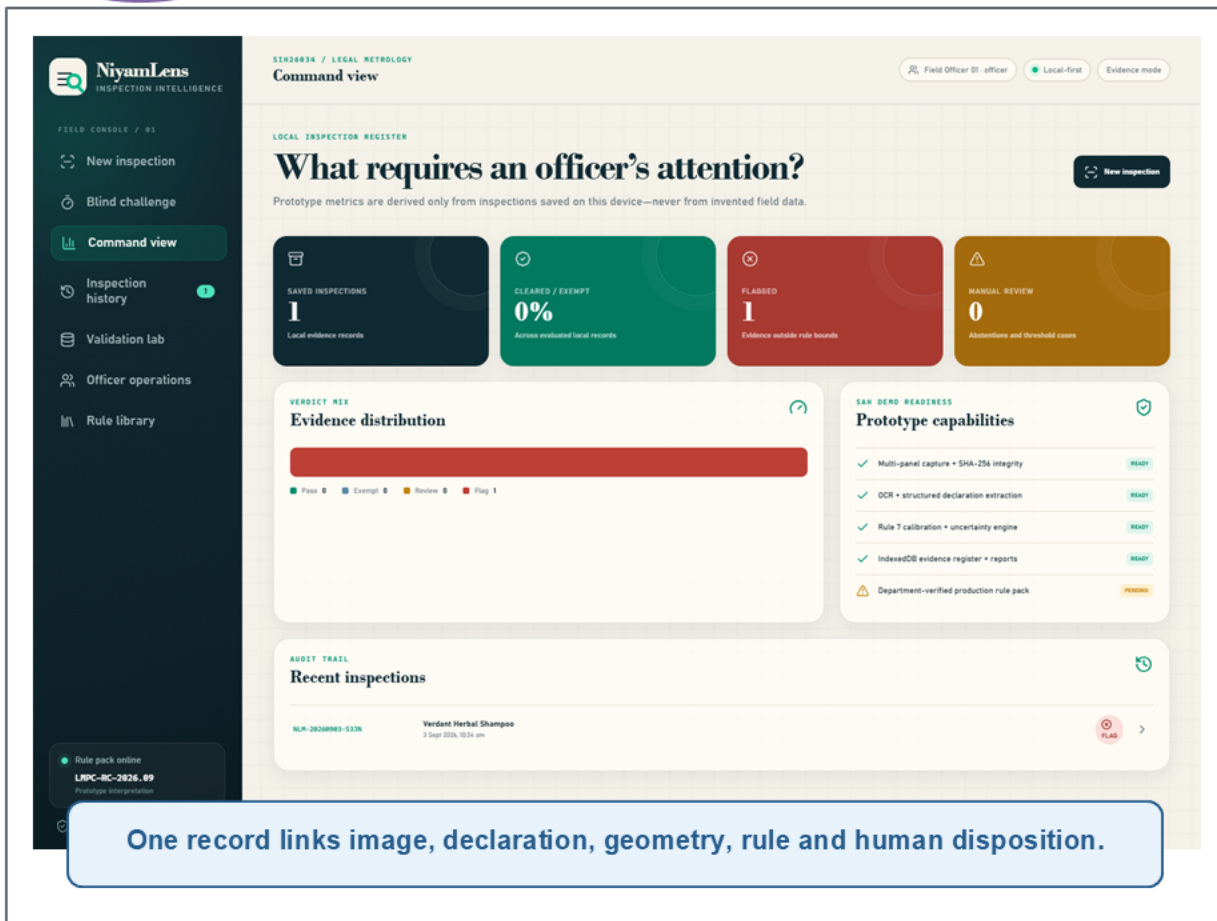
- Random-packet Blind Challenge
- Vercel-ready release candidate
- Field/legal approvals remain visible

## POTENTIAL CHALLENGES AND RISKS

- 01 Small, curved or glared text can reduce OCR recall.
- 02 Panel-area or scale error can flip a font-size result.
- 03 Exemptions and amendments can change applicability.
- 04 Inspection evidence may be offline or sensitive.
- 05 Automation could create a false accusation.

## STRATEGIES FOR OVERCOMING CHALLENGES

- 01 Quality gates, retake guidance, deep tiles and optional connected OCR.
- 02 Homography, same-panel calibration and uncertainty → REVIEW.
- 03 Versioned rule pack, Rule 26 carve-outs and legal approval register.
- 04 Local-first OCR/storage, consent and encrypted evidence transfer.
- 05 Missing view ≠ violation; preserve source, supervisor reason and audit.



## 01 INSPECTOR

### Faster triage

Guided capture, grounded fields and a one-click evidence packet.

## 02 SUPERVISOR

### Reviewable decisions

Reason-required disposition with the automated result preserved.

## 03 CONSUMER / BUSINESS

### Explainable findings

Shows what is missing, uncertain, exempt or outside the current scope.

## 04 DEPARTMENT

### Operational visibility

Local register, status dashboard and labelled validation workflow.

#### Social

Fairer, transparent inspection

#### Economic

Less repetitive manual collation

#### Operational

Auditable evidence at scale

#### Governance

Human authority stays explicit

From “trust the dashboard” to “inspect the source, rule, measurement and audit trail.”

54 / 54

AUTOMATED TESTS PASS

17

REAL PHOTOS • 10 PRODUCTS

61.1%

STANDARD TOKEN RECALL

67.5%

DEEP-SCAN TOKEN RECALL

93.7%

PRECOMPUTED CLOUD BASELINE

*Same expected tokens • no manual corrections • reliability and engine confidence are not accuracy*

## OFFICIAL / PRIMARY SOURCES

- 01 [Legal Metrology \(Packaged Commodities\) Rules, 2011 + amendments](#) — Department of Consumer Affairs
- 02 [Current Legal Metrology Acts & Rules index](#) — Department of Consumer Affairs
- 03 [Packaged Commodities FAQ](#) — Department of Consumer Affairs
- 04 [Open product-image data and API](#) — Open Food Facts
- 05 [Document text detection and request format](#) — Google Cloud Vision
- 06 [Prototype, test reports and architecture](#) — NiyamLens repository

## BEFORE ENFORCEMENT USE

- 01 Freeze a 250+ package, package-level test split with dual review.
- 02 Obtain amendment-complete Legal Metrology approval for each rule profile.
- 03 Laboratory-validate phone, marker and curved-panel measurement error.
- 04 Add departmental SSO, shared storage, retention and audit anchoring.

**WIN CONDITION:** survive the judge's unseen packet—honestly.

*Pilot measurements show progress; they are not a field-accuracy or enforcement claim.*